IF440: Artificial Intelligence

Topic 5: Bayesian Network

Moeljono Widjaja, Ph.D.
Yaya Suryana, Ph.D.

Universitas Multimedia Nusantara
Jl. Scientia Boulevard, Gading Serpong
Tangerang, Banten-15811 Indonesia

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Course Sub Learning Outcomes (Sub-CLO):

SubCLO-0221: Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing (C3).

Review

1. Using uncertain knowledge
2. Probability
3. Conditional Probability
4. Joint Probability Distribution
5. Bayes' Theorem

Outline

Pre-Class Assignments

- CS188 Spring 2014
  `https://youtu.be/gMQZq2O8yDA?t=1658&si=6oEySUf838S2GtsJ`

Probabilistic Reasoning

PROBABILISTIC REASONING

- Describe problem domain with random variables.
- Probabilistic reasoning is a method of representation of knowledge where the concept of probability is applied to indicate the uncertainty in knowledge.

PROBABILISTIC REASONING

- Examples: **Medical Diagnosis**

Symptoms and diseases can be modeled as random variables.

PROBABILISTIC REASONING

- Examples: **Business Decision Making**

Predicting customers' future behavior based on past purchasing history, important for direct marketing.

PROBABILISTIC REASONING

- Examples: **Bioinformatics**

Analysis of
Gene Expression Data

Bayesian Network

BAYESIAN NETWORK

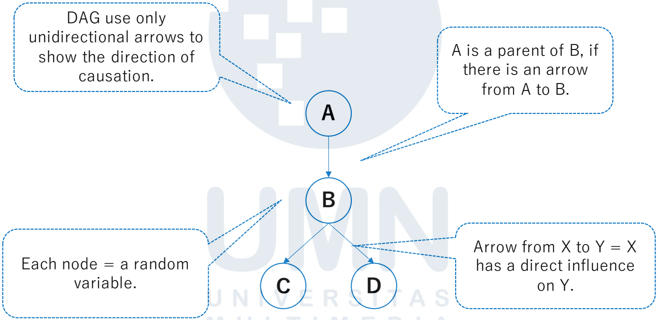
- Useful for representing and using probabilistic information.
- To do probabilistic reasoning, you need to know the joint probability distribution.
- There are two parts to any Bayesian network model:
  1) directed graph over the variables and
  2) the associated probability distribution.

BAYESIAN NETWORK

- An directed acyclic graph (DAG).
- A set of conditional probability tables (CPT) for each node in the graph.

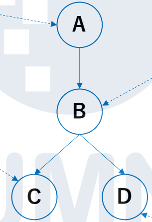
- **Syntax:**
  - **Node** = random variable (discrete or continuous).
  - **Arrows** = probabilistic dependencies between nodes.
  - Each node X_i has a **conditional probability distribution** $P(X_i | Parents(X_i))$ that quantifies the effect of the parents on the node → stored as table (CPT).

BAYESIAN NETWORK: DAG



BAYESIAN NETWORK

P(A)
0.4



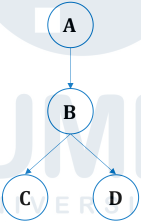
A	P(B)
false	0.01
true	0.3

B	P(C)
false	0.6
true	0.1

B	P(D)
false	0.98
true	0.95

BAYESIAN NETWORK

- $P(A=a, B=b, C=c, D=d) = P(A=a)\ P(B=b|A=a)\ P(C=c|B=b)\ P(D=d|B=b)$

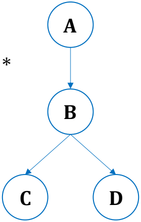


BAYESIAN NETWORK

- Suppose we want to calculate

$$P(A = true, B = true, C = true, D = true)$$
$$= P(A = true) * P(B = true \mid A = true) * P(C = true \mid B = true) *$$
$$P(D = true \mid B = true)$$ From graph structure
$$= \underbrace{0.4 * 0.3 * 0.1 * 0.95}_{\text{From CPT}}$$

$$= 0.0114$$



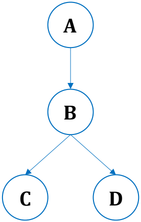
BAYESIAN NETWORK

- Suppose we want to calculate $P(B = true)$

$$= P(A = true) * P(B = true \mid A = true) + P(A = false) * P(B = true \mid A = false)$$

$$= 0.4 * 0.3 + 0.6 * 0.01$$

$$= 0.126$$



Bayesian Network Inference

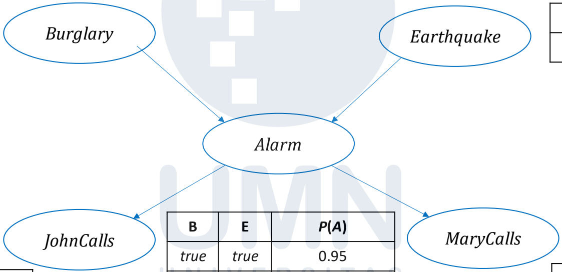
BAYESIAN NETWORK: INFERENCE

- I'm at work, neighbor John and Mary call to say my alarm is ringing. Sometimes it's set off by minor earthquakes. Is there a burglar?

- Variables: *Burglary*, *Earthquake*, *Alarm*, *JohnCalls*, *MaryCalls*.

- Network topology reflects "causal" knowledge:
  - A burglar can set the alarm off
  - An earthquake can set the alarm off
  - The alarm can cause Mary to call
  - The alarm can cause John to call

BAYESIAN NETWORK: INFERENCE



P(*B*)
0.001

P(*E*)
0.002

B	**E**	***P*(*A*)**
true	*true*	0.95
true	*false*	0.94
false	*true*	0.29
false	*false*	0.001

A	***P*(*J*)**
false	0.05
true	0.90

A	***P*(*M*)**
false	0.01
true	0.70

INFERENCE BY ENUMERATION

- In general, inference involved queries of the form:

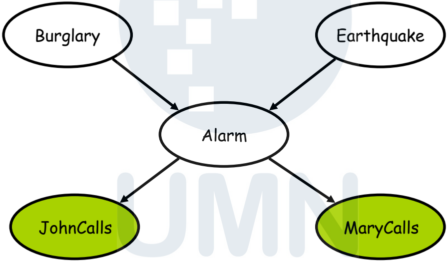
$$\mathbf{P}(X \mid e) = \alpha\ \mathbf{P}(X, e) = \alpha \sum_y \mathbf{P}(X, e, y)$$

X = the query variable(s)

e = a particular observed event

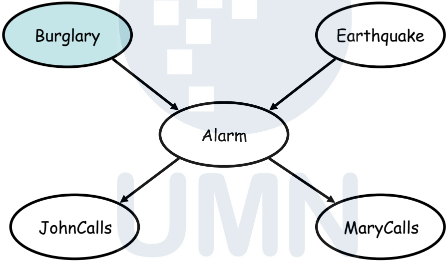
- e = the evidence variable(s)
- y = non-evidence, non-query variables (hidden variables).
- α = the normalization constant needed to make the entries in $\mathbf{P}(X| Y)$ sum to 1

Types of nodes in inference



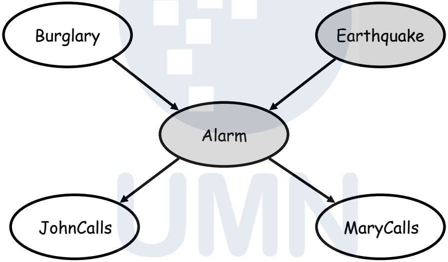
Evidence (or "observed") variables

Types of nodes in inference



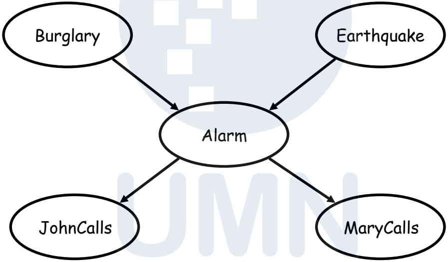
Query variables

Types of nodes in inference

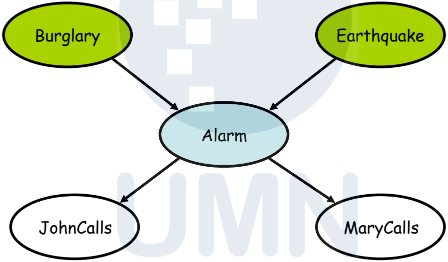


Hidden variables

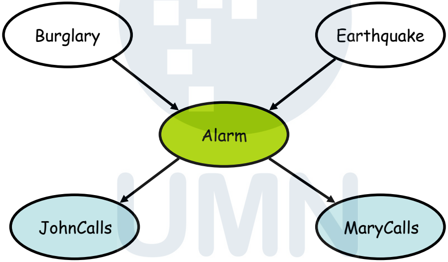
Simple inferences



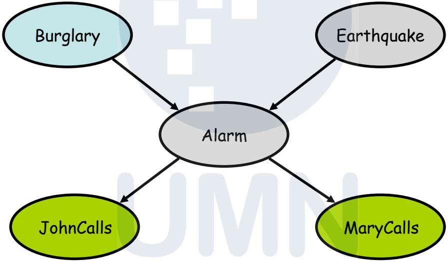
Simple inferences



Simple inferences



More difficult inferences



INFERENCE BY ENUMERATION

- Consider the query $\mathbf{P}(Burglary \mid JohnCalls{=}true, MarryCalls{=}true)$ or $\mathbf{P}(B \mid j,m)$.

$$\mathbf{P}(B \mid j, m) = \alpha\, \mathbf{P}(B,j,m) = \alpha \sum_e \sum_a \mathbf{P}(B, e, a, j, m)$$

$$P(b \mid j, m) = \alpha \sum_e \sum_a P(b)\ P(e)\ P(a \mid b, e)\ P(j \mid a)\ P(m \mid a)$$

$$P(b \mid j, m) = \alpha\, \mathrm{P}(b) \sum_e P(e) \sum_a P(a \mid b, e)\ P(j \mid a)\ P(m \mid a)$$

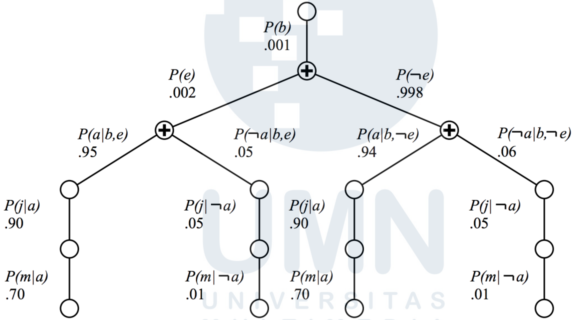
If $b{=}t,\ a{=}t,\ e{=}t$: $P(b)\ P(e)\ P(a \mid b, e)\ P(j \mid a)\ P(m \mid a)$

If $b{=}t,\ a{=}t,\ e{=}f$: $P(b)\ P(\neg e)\ P(a \mid b, \neg e)\ P(j \mid a)\ P(m \mid a)$

If $b{=}t,\ a{=}f,\ e{=}t$: $P(b)\ P(e)\ P(\neg a \mid b, e)\ P(j \mid \neg a)\ P(m \mid \neg a)$

If $b{=}t,\ a{=}f,\ e{=}f$: $P(b)\ P(\neg e)\ P(\neg a \mid b, \neg e)\ P(j \mid \neg a)\ P(m \mid \neg a)$

EVALUATION TREE



- Enumeration is inefficient: repeated computation e.g. computes $P(j|a)$ $P(m|a)$ for each value of e

INFERENCE BY ENUMERATION

We obtain

$$P(b \mid j,m) = \alpha \times 0.00059224$$

$$P(\neg b \mid j,m) = \alpha \times 0.0014919$$

Hence,

$$\mathbf{P}(B \mid j,m) = \alpha \langle 0.0059224, 0.0014919 \rangle \approx \langle 0.284, 0.716 \rangle$$

we can solve for α, which equals 1 / (P(X | e) + P(¬X | e))

That is, the chance of a burglary, given calls from both neighbors, is about 28%.

INFERENCE BY VARIABLE ELIMINATION

- Enumeration is inefficient: repeated computation e.g. computes $P(j|a)$ $P(m|a)$ for each value of e.
- Carry out summations right to left, storing intermediate results (factor) to avoid re-computation.
- Works by evaluating expressions, such as

$$P(b|j, m) = \alpha \mathrm{P}(b) \sum_e P(e) \sum_a P(a|b, e)\ P(j|a)\ P(m|a)$$

in right to left order.

INFERENCE BY VARIABLE ELIMINATION

- Consider the query $\mathbf{P}(Burglary|\ JohnCalls = true, MarryCalls = true)$.

$$\mathbf{P}(B|j, m) = \alpha\, \mathbf{P}(B) \sum_e P(e) \sum_a \mathbf{P}(a|B, e)\ P(j|a)\ P(m|a)$$

$$= \alpha\, \mathbf{f}_1(B) \sum_e \mathbf{f}_2(E) \sum_a \mathbf{f}_3(A, B, E)\ \mathbf{f}_4(A) \mathbf{f}_5(A)$$

The factors $\mathbf{f}_4(A)$ and $\mathbf{f}_5(A)$ corresponding to $P(j|a)$ and $P(m|a)$ depends on A because J and M are fixed by the query.

$$\mathbf{f}_4(A) = \begin{pmatrix} P(j|a) \\ P(j|\neg a) \end{pmatrix} = \begin{pmatrix} 0.90 \\ 0.05 \end{pmatrix} \qquad \mathbf{f}_5(A) = \begin{pmatrix} P(m|a) \\ P(m|\neg a) \end{pmatrix} = \begin{pmatrix} 0.70 \\ 0.01 \end{pmatrix}$$

INFERENCE BY VARIABLE ELIMINATION

$$\boldsymbol{f}_6(B,E) = \sum_a \mathbf{f}_3(A, B, E) \times \mathbf{f}_4(A) \times \mathbf{f}_5(A)$$

$$= (\mathbf{f}_3(a, B, E) \times \mathbf{f}_4(a) \times \mathbf{f}_5(a)) + (\mathbf{f}_3(\neg a, B, E) \times \mathbf{f}_4(\neg a) \times \mathbf{f}_5(\neg a))$$

$$= \left[\begin{pmatrix} 0.95 & 0.29 \\ 0.94 & 0.001 \end{pmatrix} x\ 0.9\ x\ 0.7\right] + \left[\begin{pmatrix} 0.05 & 0.71 \\ 0.06 & 0.999 \end{pmatrix} x\ 0.05\ x\ 0.01\right]$$

$$= \begin{pmatrix} 0.598525 & 0.183055 \\ 0.59223 & 0.0011295 \end{pmatrix}$$

$$\mathbf{P}(B|j, m) = \alpha\, \mathbf{f}_1(B) \sum_e \mathbf{f}_2(E)\ x\ \ \mathbf{f}_6(B,E)$$

INFERENCE BY VARIABLE ELIMINATION

$\mathbf{f}_7(B) = \sum_e \mathbf{f}_2(E) \times \mathbf{f}_6(B,E) = (\mathbf{f}_2(e) \times \mathbf{f}_6(B,e)) + (\mathbf{f}_2(\neg e) \times \mathbf{f}_6(B,\neg e))$

$$= \left[0.002\, x \begin{pmatrix} 0.598525 \\ 0.183055 \end{pmatrix}\right] + \left[0.998\, x \begin{pmatrix} 0.59223 \\ 0.0011295 \end{pmatrix}\right]$$

$$= \begin{pmatrix} 0.59224259 \\ 0.001493351 \end{pmatrix}$$

Finally, we get this:

$\mathbf{P}(B|j, m) = \alpha\, \mathbf{f}_1(B) \times \mathbf{f}_7(B)$

INFERENCE BY VARIABLE ELIMINATION

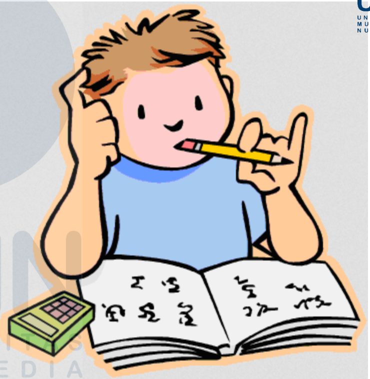
$$\mathbf{P}(B|j, m) = \alpha\, \mathbf{f}_1(B) \times \mathbf{f}_7(B)$$

$$= \alpha \begin{pmatrix} 0.001 \\ 0.999 \end{pmatrix} x \begin{pmatrix} 0.59224259 \\ 0.001493351 \end{pmatrix}$$

$$= \alpha \begin{pmatrix} 0.00059224259 \\ 0.0014918576 \end{pmatrix} \approx \langle 0.284, 0.716 \rangle$$

That is, the chance of a burglary, given calls from both neighbors, is about 28%.

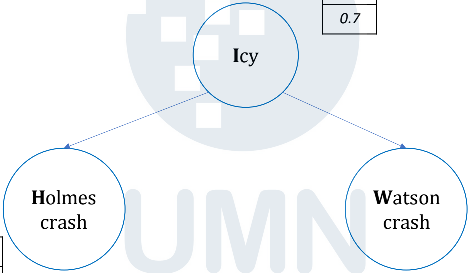
PRACTICE



PRACTICE 1

Inspector Smith is waiting for Holmes and Watson, who are driving (separately) to meet him. It is winter. His secretary tells him that Watson has had an accident. He says, “It must be that the roads are icy. I bet that Holmes will have an accident too. I should go to lunch.” But, his secretary says, “No, the roads are not icy, look at the window.” So, he says, “I guess I better wait for Holmes.”

PRACTICE 1



P(I)
0.7

I	***P*(*H*)**
t	0.8
f	0.1

I	***P*(*W*)**
t	0.8
f	0.1

PRACTICE 1

a. What is the probability that Watson crashes?

b. What is the probability of Icy given Watson crashes?

c. What is the probability of Holmes crashes given Watson crashes?

PRACTICE 2

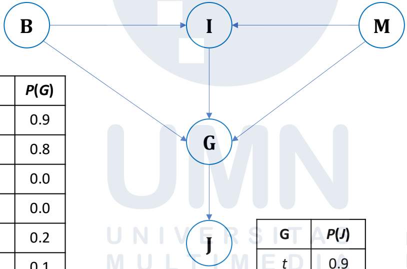
If a politician breaks election law (B), he might be indicted (I). His probability of being indicted is also influenced by whether or not the prosecuting lawyer is politically motivated (M). If he is indicted, he may be found guilty (G). The probability of being found guilty is also affected by whether he broke the law and whether his prosecutor is politically motivated. If he's found guilty, he has a high probability of going to jail (J).

PRACTICE 2

P(B)
0.9

B	M	P(I)
t	t	0.9
t	f	0.5
f	t	0.5
f	f	0.1

P(M)
0.1



B	I	M	P(G)
t	t	t	0.9
t	t	f	0.8
t	f	t	0.0
t	f	f	0.0
f	t	t	0.2
f	t	f	0.1
f	f	t	0.0
f	f	f	0.0

G	P(J)
t	0.9
f	0.0

B = BrokeElectionLaw
I = Indicted
M = PoliticallyMotivatedProsecutor
G = FoundGuilty
J = Jailed

PRACTICE 2

a. Calculate the value of $P(b, i, \neg m, g, j)$

b. Calculate the probability that someone goes to jail given that they broke the law, have been indicted, and face a politically motivated prosecutor.

c. Suppose we want to add the variable $P = PresidentialPardon$ to the network; draw the new network and briefly explain any link you add.

References

[1] S. J. Russell, P. Norvig, and E. Davis, **Artificial intelligence: a modern approach**, 3rd ed. Prentice Hall, 2010.

Next Week

- Markov Chain
- Transition Matrix

Vision and Mission

Vision

Becoming a leading Informatics Undergraduate Study Program that produces graduates with international insight who are competent in the field of computer science, have an entrepreneurial spirit, and have noble character.

Mission

1. Organising learning with the best technology and curriculum and supported by professional teaching staff.
2. Carry out research activities in the field of informatics to advance information science and technology.
3. Carry out community service activities based on information technology and science in the context of practising informatics science and technology society.